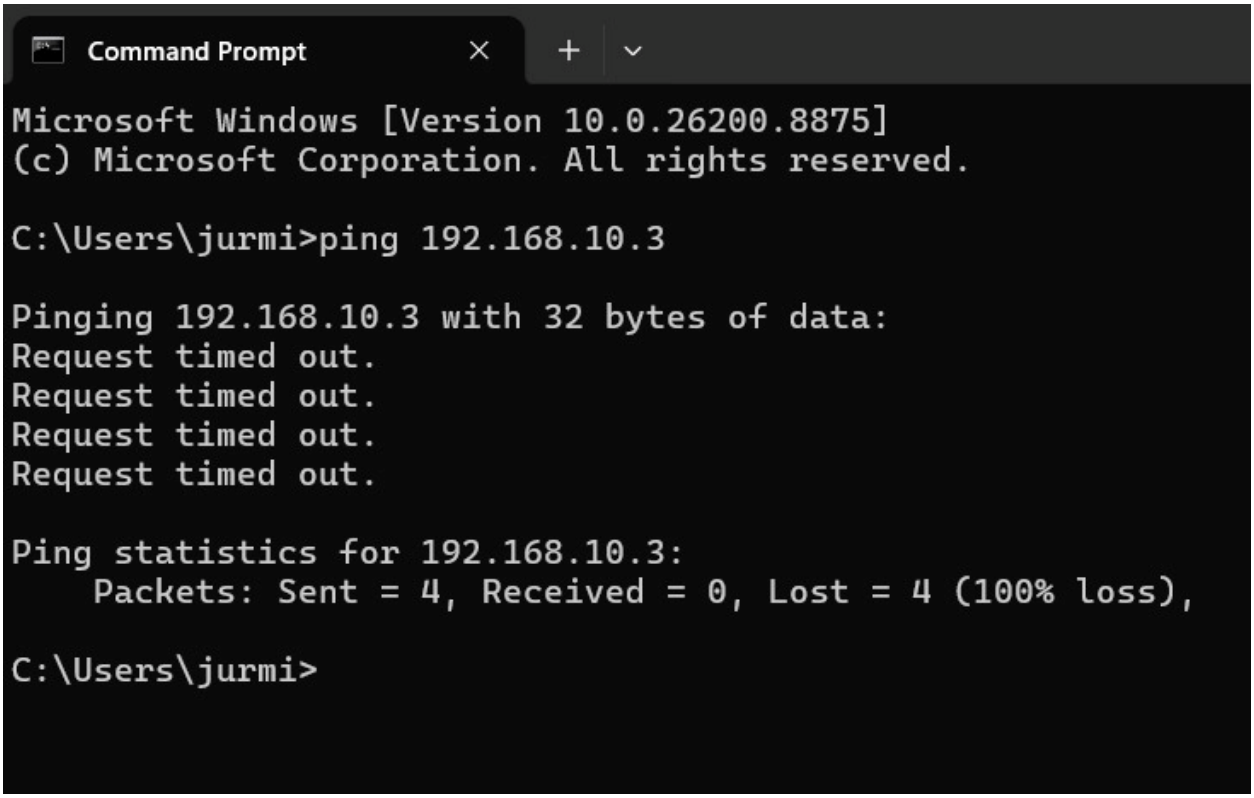
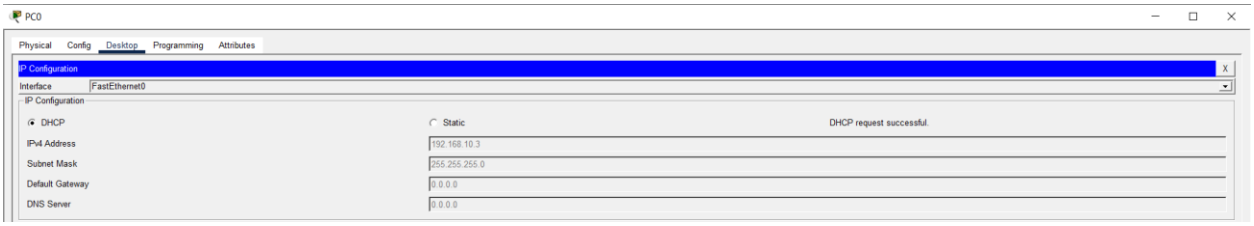
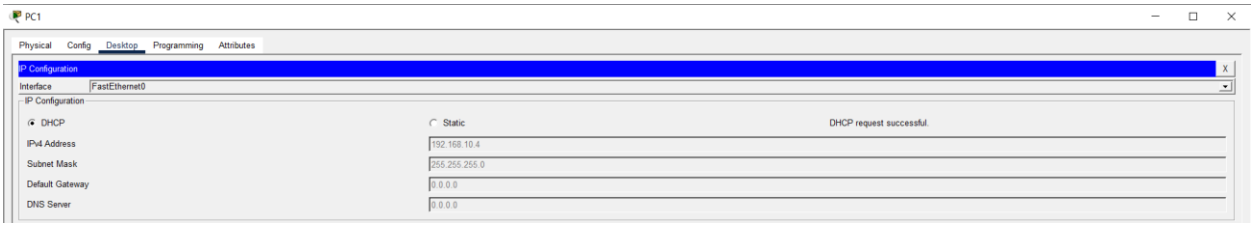
1.Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

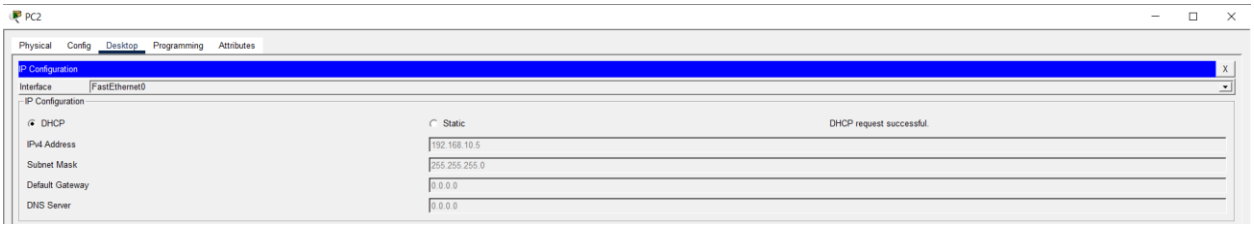


Device	IP Address	Subnet Mask
PC1	192.168.10.2	255.255.255.0
PC2	192.168.10.3	255.255.255.0
PC3	192.168.10.4	255.255.255.0
PC4	192.168.10.5	255.255.255.0
PC5	192.168.10.6	255.255.255.0

2.Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

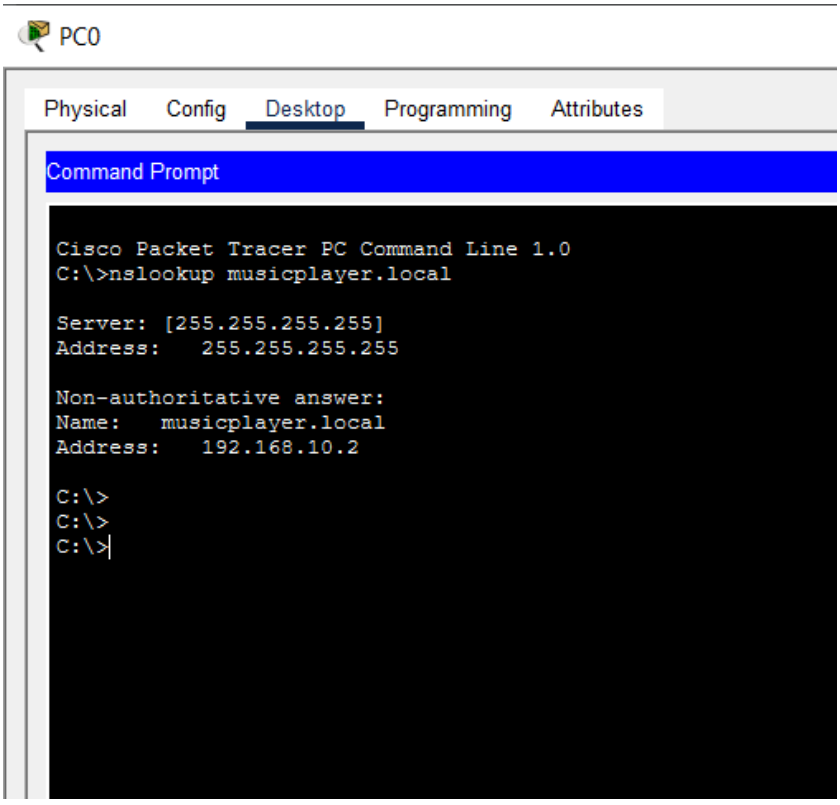






**3.Configure a DNS service (using your router’s DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

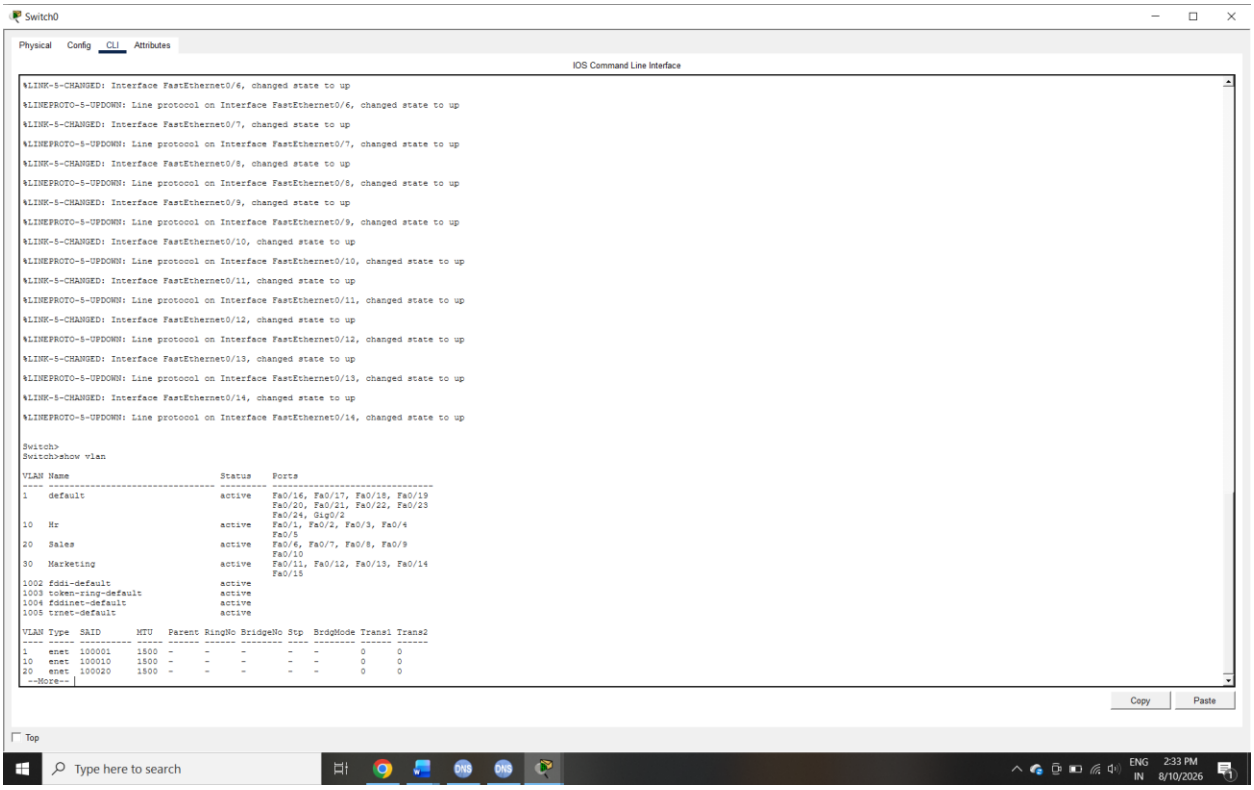
Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.**





**4.Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.**



5.Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

```
C:\Users\jurmi>ping www.wikipedia.org

Pinging dyna.wikimedia.org [2001:df2:e500:ed1a::1] with 32 bytes of data:
Reply from 2001:df2:e500:ed1a::1: time=90ms
Reply from 2001:df2:e500:ed1a::1: time=84ms
Reply from 2001:df2:e500:ed1a::1: time=83ms
Reply from 2001:df2:e500:ed1a::1: time=88ms

Ping statistics for 2001:df2:e500:ed1a::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 83ms, Maximum = 90ms, Average = 86ms

C:\Users\jurmi>
```